

**GAMIFIED VR EDUCATIONAL PLATFORM FOR
MIDDLE SCHOOL CURRICULUM IN SRILANKA -
MATHEMATICS MODULE WITH HISTORY**

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A dissertation submitted in partial fulfillment of the requirement for
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DECLARATION

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ABSTRACT

This thesis presents Lakmaga, a gamified Virtual Reality (VR) educational platform designed to teach secondary-level mathematics through the lens of ancient Sri Lankan agricultural heritage. Grounded in Situated Learning Theory, the system immerses students in a historically authentic virtual environment where mathematical concepts are embedded within culturally meaningful tasks. The platform targets four core mathematical domains: fraction arithmetic via a Sluice Gate irrigation puzzle, simultaneous equation solving via an Ancient Merchant's Balance Scale task, volume calculation via a Pit Excavation challenge, and physics-based lever mechanics. Developed using Unity 2022 LTS with Meta Quest 3 as the target hardware, the system employs XR Interaction Toolkit for natural hand-based interactions, a stealth assessment engine with CSV-based telemetry logging, and an adaptive achievement system with three-tier medal evaluation (Apprentice, Village Master, Royal Engineer). Preliminary evaluation with a cohort of 30 participants demonstrated a 90% task accuracy rate, stable frame rates averaging 72 fps on Meta Quest 3 hardware, and strong positive user engagement scores. The findings suggest that culturally contextualised VR gamification can significantly enhance mathematical understanding while preserving and transmitting intangible cultural heritage to younger generations.

Keywords: Virtual Reality, Gamified Education, Mathematics Learning, Situated Learning Theory, Cultural Heritage Preservation

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LIST OF ABBREVIATIONS

VR	Virtual Reality
XR	Extended Reality
UI	User Interface
UX	User Experience
CSV	Comma-Separated Values
FPS	Frames Per Second
SDK	Software Development Kit
API	Application Programming Interface
SUS	System Usability Scale
SLIIT	Sri Lanka Institute of Information Technology
LTS	Long-Term Support
TMP	TextMeshPro

CHAPTER 1: INTRODUCTION

1.1 Background

The integration of Virtual Reality (VR) technology into educational settings represents one of the most promising developments in contemporary pedagogy. As traditional classroom-based instruction struggles to maintain student engagement in an increasingly digital world, immersive technologies offer a compelling alternative that leverages the principles of experiential learning. VR enables students to interact with abstract concepts in tangible, three-dimensional environments, transforming passive reception of information into active, embodied learning experiences. This paradigm shift is particularly relevant for mathematics education, where the abstract nature of concepts such as fractions, algebraic equations, and volumetric calculations often poses significant barriers to student comprehension and retention.

Sri Lanka possesses a remarkably rich heritage of agricultural engineering that spans over two millennia. Ancient Sinhalese civilizations developed sophisticated irrigation systems, including the world-renowned Tank Cascade Systems (Wewa Wari Samuhaya), the Bisokotuwa (valve pit) pressure management technology, and intricate sluice gate mechanisms that demonstrate an advanced understanding of hydraulics, mathematics, and material science. These innovations, many of which predate similar Western technologies by centuries, represent an invaluable repository of applied mathematical and engineering knowledge that remains largely untapped as an educational resource.

The Lakmaga project (from Sinhala: ලක්මග, meaning 'Path of Lanka') was conceived to bridge this gap by creating a gamified VR educational platform that situates mathematics learning within the authentic context of ancient Sri Lankan agricultural heritage. The platform targets secondary school students (Grades 6-12) and employs Situated Learning Theory as its theoretical foundation, embedding mathematical problem-solving within culturally meaningful, historically grounded tasks that give abstract concepts concrete, relatable meaning.

The platform is developed using Unity 2022 LTS with C# scripting, targeting the Meta Quest 3 standalone VR headset. It utilizes the XR Interaction Toolkit for natural hand-based interactions, TextMeshPro for VR-optimized text rendering, and a custom-built telemetry engine for stealth performance assessment. The system architecture follows a modular design pattern with singleton managers, enabling independent development and testing of educational modules while maintaining a cohesive gameplay experience.

This thesis specifically focuses on the Mathematics Modules of the Lakmaga platform, encompassing three primary interactive tasks: (1) the Sluice Gate Fractions Task, which teaches fraction arithmetic through water level management; (2) the Ancient Merchant's Balance Scale Task, which teaches simultaneous equation solving through physical weight balancing; and (3) the Pit Excavation Volume Challenge, which teaches volume calculation and division through construction planning. Each task is designed to align with the Sri Lankan secondary mathematics curriculum while providing an engaging, culturally enriched learning experience.

1.2 Research Gap

Despite the growing body of research on VR-based education, several critical gaps persist in the current literature and practice. First, the majority of existing VR educational applications focus on STEM subjects within Western cultural contexts, with minimal representation of South Asian or specifically Sri Lankan heritage. This cultural homogeneity limits the relevance and engagement potential of these tools for diverse student populations, particularly those in Sri Lanka who may benefit most from culturally familiar learning contexts.

Second, while numerous VR mathematics applications exist, they predominantly adopt a direct instruction approach, presenting mathematical problems in virtual whiteboards or abstract puzzle environments that fail to leverage the immersive potential of VR. Few applications successfully embed mathematical concepts within authentic, narrative-driven scenarios that provide genuine contextual meaning to abstract operations. The

disconnect between mathematical abstraction and real-world application remains a persistent challenge.

Third, the assessment mechanisms in existing VR educational tools are often rudimentary, relying on simple pass/fail evaluations rather than comprehensive, multi-dimensional performance tracking. There is a notable absence of stealth assessment approaches that capture nuanced behavioral data including time-on-task, error patterns, interaction sequences, and adaptive difficulty responses without disrupting the immersive experience.

Fourth, gamification in VR education has been largely limited to superficial reward systems such as points and badges, without deep integration of game mechanics into the pedagogical design. The potential for physics-based puzzles, tactile manipulation, and embodied problem-solving to serve as both engagement mechanisms and assessment instruments remains underexplored.

Finally, there is a significant gap in leveraging intangible cultural heritage as an educational scaffold. While UNESCO recognizes the importance of preserving traditional knowledge systems, few technology-enhanced learning platforms have attempted to use ancient engineering achievements as vehicles for teaching modern curriculum-aligned mathematics.

1.3 Research Problem

The central research problem addressed by this thesis is: How can a gamified Virtual Reality platform effectively teach secondary-level mathematical concepts by contextualizing them within authentic ancient Sri Lankan agricultural heritage tasks, while maintaining measurable learning outcomes through integrated stealth assessment?

This problem encompasses several interrelated sub-problems: (a) the design of VR interactions that faithfully represent historical engineering mechanisms while embedding curriculum-aligned mathematical operations; (b) the development of physics-based puzzle mechanics that require genuine mathematical reasoning rather than trial-and-error

gameplay; (c) the implementation of unobtrusive performance tracking that captures meaningful learning analytics without disrupting immersion; and (d) the evaluation of learning effectiveness and user experience in the target demographic.

1.4 Research Objectives

The primary objectives of this research are as follows:

1. To design and develop a VR-based mathematics learning module that teaches fraction arithmetic through an interactive sluice gate irrigation simulation, incorporating rotational snapping physics, equivalent fraction visualization, and water volume control mechanics.
2. To design and develop a VR-based simultaneous equation solving module using an ancient merchant's balance scale, incorporating dynamic weight calculations, collider-based mass detection, equilibrium color triggers, and algebraic substitution mechanics.
3. To implement a stealth assessment engine with CSV-based telemetry logging that captures task start/completion times, action sequences, error counts, and quiz performance without disrupting the VR experience.
4. To integrate a three-tier gamification system (Apprentice, Village Master, Royal Engineer) with adaptive evaluation criteria based on time and error metrics.
5. To evaluate the system's effectiveness through user testing, measuring task accuracy, completion times, frame rate stability, and user satisfaction.

1.5 Scope and Limitations

The scope of this research is limited to the Mathematics Modules of the Lakmaga platform, specifically the Sluice Gate Fractions Task, the Merchant Balance Scale Task, and the Pit Excavation Volume Challenge. While the broader Lakmaga platform encompasses additional educational domains including science, engineering, and cultural heritage modules developed by other team members, this thesis focuses exclusively on the mathematical components.

The target hardware platform is the Meta Quest 3 standalone VR headset, and all performance metrics are reported for this specific device. The evaluation cohort consists of 30 participants, which, while sufficient for preliminary validation, limits the generalizability of the findings. The system operates entirely offline, storing telemetry data locally on the device, which eliminates cloud dependency but limits real-time teacher monitoring capabilities.

1.6 Structure of the Thesis

This thesis is organized into four chapters. Chapter 1 introduces the research context, identifies the research gap and problem, and states the objectives. Chapter 2 presents the methodology, including detailed technical descriptions of each mathematical task's implementation based on the Unity C# codebase, the telemetry system, and the testing approach. Chapter 3 reports the evaluation results, discusses findings, and summarizes individual contributions. Chapter 4 concludes the thesis with a summary of achievements, limitations, and directions for future work.

CHAPTER 2: METHODOLOGY

2.1 Research methodology

This research adopts a Design Science Research (DSR) methodology, which is particularly well-suited for information systems research that involves the creation and evaluation of IT artifacts. The DSR framework comprises iterative cycles of problem identification, solution design, development, demonstration, and evaluation. The methodology enables a rigorous approach to building the Lakmaga VR platform while simultaneously generating theoretical contributions to the fields of VR-based education and cultural heritage computing.

The development process followed an Agile-inspired iterative approach with two-week sprint cycles. Each sprint focused on a specific educational task or system component, progressing through requirements analysis, design, implementation, integration testing, and user feedback. This iterative structure allowed continuous refinement of both the pedagogical design and the technical implementation based on formative evaluation data.

The theoretical foundation of the pedagogical design is Situated Learning Theory (Lave and Wenger, 1991), which posits that learning is most effective when it occurs within authentic contexts that mirror the real-world application of knowledge. By embedding mathematical tasks within historically accurate simulations of ancient Sri Lankan agricultural practices, the platform provides situational context that transforms abstract mathematical operations into meaningful, purpose-driven problem-solving activities. The VR medium further enhances this situated approach by enabling embodied cognition, where physical interactions with virtual objects reinforce conceptual understanding through sensorimotor engagement.

Data collection employed a mixed-methods approach combining quantitative telemetry data (automatically captured by the ProgressTracker system) with qualitative user feedback gathered through post-session questionnaires and semi-structured interviews. Quantitative metrics include task accuracy rates, completion times, error frequencies,

and quiz scores. Qualitative data capture user perceptions of engagement, cultural relevance, and learning effectiveness.

2.2 System architecture

The Lakmaga platform is architected as a modular Unity application comprising several interconnected subsystems. The core architecture follows a singleton manager pattern, where each major system component is implemented as a persistent singleton `MonoBehaviour` that survives scene transitions via `DontDestroyOnLoad`. This design ensures consistent state management across the multi-scene VR experience.

The primary architectural components are organized into four layers:

Presentation Layer: Comprises all VR user interface elements built using Unity's World Space Canvas system with `TrackedDeviceGraphicRaycaster` for XR controller interaction. UI panels are constructed programmatically at runtime using `TextMeshPro` for high-quality text rendering with outline effects for VR readability. Key UI classes include `SluiceGatePuzzleUI`, `MerchantBalanceUI`, and `MathChallengeTablet`.

Interaction Layer: Utilizes the XR Interaction Toolkit with `XRGrabInteractable` components for natural hand-based object manipulation. Objects such as `WeighableSack` and `StandardWeight` implement custom grab behaviors with visual feedback (color highlights on hover and grab states), haptic responses, and audio cues. The `SluiceGateWheel` implements a custom rotational interaction system that tracks hand angular displacement to control wheel rotation.

Logic Layer: Contains the core puzzle controllers (`SluiceGateController`, `MerchantBalanceManager`, `MathChallengeTablet`, `BalanceScaleController`) that manage game state, evaluate player answers, and coordinate between UI and interaction components. Each controller implements a finite state machine pattern to manage puzzle progression.

Data Layer: Encompasses the `ProgressTracker` (CSV telemetry), `SaveManager` (JSON player profiles via `PlayerData`), `AchievementManager` (medal evaluation), and

VRQuizManager (stealth assessment). All data persistence uses Application.persistentDataPath for Meta Quest compatibility, requiring no internet connectivity.

2.3 Mathematics module design

The Mathematics Modules are designed around three core principles derived from Situated Learning Theory and gamification research:

Principle 1 - Authentic Context: Each mathematical task is embedded within a historically accurate simulation of an ancient Sri Lankan agricultural practice. The Sluice Gate task simulates water management in the Horowpotana irrigation system; the Merchant Balance task simulates spice trade weighing practices; and the Pit Excavation task simulates construction planning for ancient granary foundations.

Principle 2 - Embodied Interaction: Mathematical operations are performed through physical VR interactions rather than abstract symbol manipulation. Students turn wheels to set fraction values, physically place weighted objects on scale pans, and enter calculations through VR-compatible number pad interfaces. This embodied approach leverages the motor-cognitive coupling described in embodied cognition theory.

Principle 3 - Integrated Assessment: Performance evaluation is woven into the gameplay through stealth assessment mechanisms. The ProgressTracker continuously logs timestamped actions and errors to CSV files, while the VRQuizManager presents contextual multiple-choice questions at task completion gates. The AchievementManager synthesizes time and error data into a three-tier medal system that provides immediate, motivating feedback.

2.4 Sluice gate fractions task

2.4.1 Educational objectives

The Sluice Gate Fractions Task is designed to teach fraction arithmetic, specifically fraction subtraction, equivalent fraction recognition, and decimal-fraction conversion. The learning scenario presents students with a water tank at $\frac{3}{4}$ ($\frac{6}{8}$) capacity and

challenges them to release water until exactly $\frac{3}{8}$ remains, requiring the calculation: $\frac{6}{8} - \frac{3}{8} = \frac{3}{8}$ of water to release.

2.4.2 Core controller implementation (SluiceGateController.cs)

The SluiceGateController class forms the mathematical backbone of the puzzle. It defines nine fraction constants as public static floats, enabling precise positional mapping between decimal values and their fractional representations. The implementation defines:

```
public const float FRACTION_0_8 = 0f;    // 0/8 = Empty
public const float FRACTION_1_8 = 0.125f; // 1/8
public const float FRACTION_2_8 = 0.25f;  // 2/8 = 1/4
public const float FRACTION_3_8 = 0.375f; // 3/8 = TARGET
public const float FRACTION_4_8 = 0.5f;   // 4/8 = 1/2
public const float FRACTION_5_8 = 0.625f; // 5/8
public const float FRACTION_6_8 = 0.75f;  // 6/8 = 3/4 = START
public const float FRACTION_7_8 = 0.875f; // 7/8
public const float FRACTION_8_8 = 1f;     // 8/8 = Full
```

The controller initializes with startingWaterLevel set to FRACTION_6_8 (0.75, representing $\frac{3}{4}$) and targetWaterLevel set to FRACTION_3_8 (0.375, representing $\frac{3}{8}$). A configurable tolerance of 0.05 provides ergonomic flexibility for VR interactions, allowing answers within plus or minus 5% of the exact target to be accepted as correct.

The OnWheelValueChanged method receives continuous input from the SluiceGateWheel and clamps the value to the [0,1] range. It updates the currentWaterLevel, triggers visual updates through the WaterLevelVisualizer, and plays spatial audio feedback through a 3D AudioSource. The method also fires the OnWheelChanged UnityEvent, enabling decoupled communication with other scene components.

2.4.3 Answer evaluation logic

The EvaluateAnswer method implements a three-outcome evaluation system. When the player confirms their answer, the method calculates the absolute difference between the

wheel position and the target water level. If the difference falls within the tolerance threshold, a SUCCESS outcome is triggered: the ProgressTracker logs the correct fraction, the SluiceGateFeedback plays a success voice line with culturally appropriate Sinhala narration, and the VRQuizManager intercepts to present a stealth assessment quiz before firing the OnPuzzleSuccess event.

If the wheel position is below the target minus tolerance (too much water released), a FAILURE_TOO_MUCH outcome triggers flooding visual effects, drought particle systems, and a Sinhala narrator voice line warning of field flooding. If the wheel position is above the target plus tolerance (insufficient water released), a FAILURE_NOT_ENOUGH outcome triggers dried field visuals and an appropriate warning narration. Both failure outcomes increment the errorCount and log the incorrect fraction attempt to the ProgressTracker.

2.4.4 Rotational snapping physics (SnapToNearestNotch)

A critical UX feature is the SnapToNearestNotch method, which provides tactile precision for VR interactions. The method defines an array of nine notch positions corresponding to the fraction constants (0/8 through 8/8). When invoked, it iterates through all notch positions, calculates the absolute distance from the current wheel position to each notch, and snaps the wheel to the nearest notch. This discrete snapping mechanism transforms the continuous wheel rotation into a set of meaningful fractional positions, reinforcing the mathematical concept that fractions represent discrete subdivisions of a whole.

2.4.5 VR wheel interaction (SluiceGateWheel.cs)

The SluiceGateWheel class implements a physically intuitive VR rotation mechanic. The wheel is configured with an XRGrabInteractable component with trackPosition and trackRotation both disabled, preventing the wheel from following the hand position and instead allowing manual rotation calculation. When the player grabs the wheel, OnSelectEntered records the initial hand angle relative to the wheel center using Atan2 in the local X-Y plane. During Update, CalculateRotationsFromHand computes the

angular delta between the current hand position and the grab start angle, applies this delta to the wheel visual's local Z-axis euler angle, and accumulates the total rotation.

The wheel maps rotations to three discrete water levels: 1 rotation maps to lowWaterLevel ($1/8 = 0.125$), 2 rotations to perfectWaterLevel ($3/8 = 0.375$, the correct answer), and 3 rotations to tooMuchWaterLevel ($5/8 = 0.625$). Upon release (OnSelectExited), the LockInRotation method sets the water level based on the current rotation count and communicates the result to both the SluiceGateController and the RiceFieldIrrigationController, which handles the visual water flow and rice plant growth animations.

2.4.6 Visual feedback system (SluiceGateFeedback.cs)

The SluiceGateFeedback class orchestrates a multi-sensory feedback sequence using Unity coroutines. Each feedback type (Success, FailureTooMuch, FailureNotEnough) triggers a coroutine that: (1) waits for a configurable delay, (2) activates the narrator's talking animation via Animator.SetBool, (3) plays the corresponding visual effects (water flow particles for success, flooding particles for excess water, drought particles for insufficient water), (4) plays a spatialized voice line through either an external AudioSource on the farmer character or a local AudioSource, (5) displays subtitles through the DialogueUI system, and (6) enables/disables environment GameObjects to show green healthy fields, flooded fields, or dried brown fields. The 3D spatial audio (spatialBlend = 1f) ensures that the farmer's voice emanates from the character's position in the VR scene.

2.4.7 Water level visualization (WaterLevelVisualizer.cs)

The WaterLevelVisualizer provides real-time visual representation of the water level using both 3D scene elements and 2D UI. The water surface Transform is vertically positioned between configurable minWaterHeight and maxWaterHeight values based on the current display level. A critical design feature is the anti-decrease mechanism: the visualizer tracks the highest level reached (m_HighestLevel) and prevents the display from decreasing below this value, ensuring water only fills up during irrigation mode.

The SetWaterLevel method only accepts increases, while ResetWaterLevel provides the only path to decrease the level, maintaining physical realism.

Color-coded feedback changes the water material color: normalWaterColor (blue, RGBA 0.2, 0.6, 1.0, 0.8), successColor (green), dangerLowColor (red for flooding), and dangerHighColor (dark red for drought). The fraction text display uses the static GetFractionString method from SluiceGateController to show human-readable fraction labels such as '3/8' or '1/4 (2/8)', reinforcing equivalent fraction understanding.

2.4.8 Puzzle UI panel (SluiceGatePuzzleUI.cs)

The SluiceGatePuzzleUI creates a world-space Canvas panel sized at 2.0m x 1.6m for VR readability, with TrackedDeviceGraphicRaycaster enabling XR controller interaction. The panel displays the problem statement with color-coded fraction values (3/4 in gold, 3/8 in green), a tank diagram with animated water fill, a target line at the 3/8 position, fraction markers at each eighth increment, and a Quick Reference card showing fraction-to-decimal conversions. The panel smoothly faces the camera using Quaternion.Slerp at 3x Time.deltaTime speed. The status text dynamically updates with color changes based on proximity to the target: green when within tolerance, yellow when close, and white otherwise.

2.5 Merchant balance scale task

2.5.1 Educational objectives

The Merchant Balance Scale Task teaches simultaneous equation solving and algebraic substitution. Students are presented with two simultaneous linear equations where x represents the weight of a Cinnamon sack and y represents the weight of a Pepper sack. They must physically balance each equation using the VR scale, then solve the system algebraically to determine the individual weights.

2.5.2 Puzzle generation system (MerchantBalanceManager.cs)

The MerchantBalanceManager implements a difficulty-tiered puzzle generation system with predefined presets that guarantee clean integer or simple decimal solutions. Three difficulty levels are supported:

Easy: Puzzles with small coefficients (1-3) and integer solutions. Example: Equation 1: $2x + 1y = 7\text{kg}$; Equation 2: $1x + 1y = 5\text{kg}$; Solution: $x = 2\text{kg}$, $y = 3\text{kg}$. Three easy presets are defined.

Medium: Puzzles with larger coefficients (1-3) and potentially non-integer solutions. Example: Equation 1: $3x + 2y = 16\text{kg}$; Equation 2: $1x + 2y = 10\text{kg}$; Solution: $x = 3\text{kg}$, $y = 3.5\text{kg}$. Two medium presets are defined.

Hard: Puzzles with coefficients up to 5 and larger total weights. Example: Equation 1: $3x + 4y = 26\text{kg}$; Equation 2: $2x + 3y = 19\text{kg}$; Solution: $x = 2\text{kg}$, $y = 5\text{kg}$. Two hard presets are defined.

The GeneratePuzzle method randomly selects a preset from the appropriate difficulty tier. Each PuzzlePreset contains two EquationData objects (storing cinnamonCount, pepperCount, and totalWeight) and the solution values (cinnamonWeight and pepperWeight). The EquationData class includes utility methods ToEquationString() and ToDescriptionString() for generating human-readable equation representations in the UI.

2.5.3 Puzzle state machine

The puzzle progression follows an eight-state finite state machine: NotStarted, Equation1Active, Equation1Balanced, Equation2Active, Equation2Balanced, AwaitingAnswer, Solved, and Failed. Transitions are triggered by scale balance events (OnScaleBalanced, OnScaleUnbalanced) and player actions (ProceedToEquation2, SubmitAnswer). The state machine ensures that players must physically balance each equation before proceeding, and both equations must be balanced before the answer input panel becomes accessible.

2.5.4 Object spawning system

The SpawnEquationItems method creates WeighableSack objects for each equation's required spice sacks. Sacks are positioned at the sackSpawnPoint with 0.35m horizontal spacing between each sack and slight random Z-axis variation for visual naturalness. The SpawnStandardWeights method creates two copies of each available weight value (1kg, 2kg, 3kg, 5kg) at the weightSpawnPoint with 0.25m spacing. Both methods support either prefab-based instantiation (if sackPrefab or weightPrefab are assigned) or automatic primitive creation (Capsule for sacks, Cube for weights) with appropriate scaling.

2.5.5 Balance scale physics (BalanceScaleController.cs)

The BalanceScaleController implements a physics-based beam balance simulation. The controller references two ScalePan components (left and right) and a beam pivot Transform. Weight changes are received via UnityEvent callbacks from the ScalePan components. The tilt calculation normalizes the weight difference (leftWeight - rightWeight) against the maximum combined weight to produce a value in the [-1, 1] range, which is then scaled by the maxTiltAngle (default 25 degrees) to produce the target tilt angle. Smooth interpolation using Mathf.Lerp with tiltSpeed (default 3) produces physically plausible beam motion.

Equilibrium detection uses a configurable balanceTolerance (default 0.1kg). The scale is considered balanced when the absolute weight difference is within tolerance and the

combined weight is greater than zero. State transitions between balanced and unbalanced states trigger `OnBalanceReached` and `OnBalanceLost` UnityEvents respectively, with equilibrium chime audio and creak sounds during tilting. Visual feedback changes the beam renderer color between `balancedColor` (green) and `unbalancedColor` (amber) using `Color.Lerp` for smooth transitions. The pan transforms move vertically based on tilt, with the heavier side descending by `panVerticalRange` (default 0.3m).

2.5.6 Collider-based weight detection (ScalePan.cs)

Each `ScalePan` uses a trigger `BoxCollider` (default size 0.6m x 0.3m x 0.6m) to detect objects placed on it. The `OnTriggerEnter` method checks for `WeighableSack` or `StandardWeight` components on entering colliders. When a weighable object is detected, it is added to the `itemsOnPan` list, and `RecalculateWeight` iterates through all items, summing `GetWeight()` for sacks and `weightValue` for standard weights. The `SnapItemToPan` method parents the object to the pan, sets its `Rigidbody` to kinematic, and positions it in a grid layout with `stackHeightOffset` (0.05m) between stacked items. When objects exit the trigger, `ReleaseItemFromPan` un-parents them and restores non-kinematic physics.

2.5.7 Weighable objects (WeighableSack.cs, StandardWeight.cs)

`WeighableSack` implements a VR-grabbable spice sack with `XRGrabInteractable` configured for dynamic attach (`useDynamicAttach = true`) and no throw on detach (`throwOnDetach = false`). Each sack has a `hiddenWeight` (set by the `MerchantBalanceManager`, not visible to the player) and a `spiceType` (`Cinnamon` or `Pepper`). Visual differentiation uses color coding: `cinnamonColor` (brown, RGB 0.55, 0.27, 0.07) and `pepperColor` (red, RGB 0.7, 0.15, 0.15). 3D `TextMeshPro` labels display the spice type and variable name ('x' for Cinnamon, 'y' for Pepper) on both the front and top surfaces. Hover and grab states apply distinct highlight colors for immediate visual feedback.

`StandardWeight` implements known-mass iron calibration weights with visible weight labels. Unlike sacks, the weight value is prominently displayed on the object's top and

front faces using TextMeshPro. The cube size scales with weight value using the formula: $\text{scaleFactor} = \text{Clamp}(\text{weightValue} * 0.03 + 0.08, 0.08, 0.25)$, providing intuitive visual mass indication.

2.5.8 Answer submission and validation

The MerchantBalanceUI provides a VR number pad interface for entering x and y values. The number pad features buttons 0-9 arranged in two rows, with a Clear button and Submit button. Players toggle between x (Cinnamon) and y (Pepper) input fields using dedicated selection buttons. The SubmitAnswer method validates both values against the puzzle solution with a tolerance of 0.1kg. Correct answers trigger the Solved state with celebratory feedback and a stealth quiz via VRQuizManager. Incorrect answers provide specific feedback indicating which variable is wrong, increment the errorCount, and log the attempt to ProgressTracker.

2.6 Pit excavation volume task

2.6.1 Educational objectives

The Pit Excavation Volume Challenge teaches volume calculation ($V = L \times W \times H$) and integer division. Students calculate the volume of a pit layer and divide by the cartload volume to determine the number of cartloads needed to fill the pit with rubble for granary foundation construction.

2.6.2 Implementation (MathChallengeTablet.cs)

The MathChallengeTablet presents the problem on a VR-friendly world-space tablet. The problem is dynamically generated from configurable parameters: pitLength (default 5m), pitWidth (default 5m), layerHeight (default 0.4m), and cartloadVolume (default 2 cubic metres). The correct answer is computed at runtime: $\text{Volume} = \text{pitLength} \times \text{pitWidth} \times \text{layerHeight} = 10 \text{ cubic metres}$; $\text{Cartloads} = \text{Ceiling}(\text{Volume} / \text{cartloadVolume}) = 5 \text{ cartloads}$.

The UI features a VR number pad with buttons 0-9, a Clear button (C), and a Submit button (checkmark). The CheckAnswer method compares the player's integer input

against the computed correctAnswer. Correct answers unlock the stone MaterialPile for the next construction phase and transition the UI to show the ExcavatedPitGuidePanel. Wrong answers display the hint 'Volume = L x W x H' and clear the input after 1.5 seconds.

2.7 Telemetry and assessment engine

2.7.1 ProgressTracker system

The ProgressTracker is a singleton MonoBehaviour that logs student performance metrics to a local CSV file stored at Application.persistentDataPath, ensuring offline compatibility with the Meta Quest 3 standalone headset. The CSV schema consists of five columns: Timestamp (yyyy-MM-dd HH:mm:ss format), TaskName (string identifier), ActionDetail (descriptive string), IsError (boolean flag), and TimeTaken (float in seconds, populated only for TASK_COMPLETED events).

Performance optimization is achieved through a StringBuilder buffer that accumulates log entries in memory, flushing to disk only on task completion (via LogTaskComplete) or at configurable intervals (flushIntervalSeconds, default 10 seconds). This design prevents per-frame file I/O operations that could cause frame drops on the Quest hardware. Additional safety flushes occur on OnApplicationPause (when the headset is removed), OnApplicationQuit, and OnDestroy, preventing data loss during unexpected session termination. The EscapeCsv method handles comma and quote escaping for CSV format integrity.

2.7.2 VRQuizManager stealth assessment

The VRQuizManager implements a stealth assessment pattern where multiple-choice quizzes are presented at task completion gates, appearing as natural game progression rather than explicit tests. The system maintains a registry of QuizData ScriptableObject

assets, each containing a syllabusTopic identifier and a list of QuizQuestion objects with question text, four answer options, and the correct answer index.

When ShowQuiz is called with a topic string and success callback, the manager selects a random question from the matching QuizData, spawns a world-space UI panel 2 metres in front of the player camera, and presents the four answer options as interactive buttons with colored accent strips. Correct answers display green feedback and invoke the success callback after a 1.5-second delay. Incorrect answers display red feedback, disable the selected button to prevent re-selection, and log the error attempt. The quiz UI billboards toward the player using LateUpdate rotation tracking.

2.7.3 Achievement and gamification system

The AchievementManager evaluates student performance using a three-tier medal system inspired by the cultural hierarchy of ancient Sri Lankan craftspeople:

Royal Engineer (Gold) - Awarded for 0 errors and completion within the fast time threshold (default 60 seconds, configurable per task). Awards 100 points.

Village Master (Silver) - Awarded for fewer than 3 errors regardless of time. Awards 75 points.

Apprentice (Bronze) - Awarded for 3 or more errors. Awards 50 points, ensuring all participants receive positive reinforcement.

The EvaluatePerformance method receives the taskName, timeTaken, and errorCount, determines the appropriate medal tier, and calls SaveAchievementToProfile to persist the achievement to the PlayerData JSON structure via the SaveManager. Visual feedback includes a confetti ParticleSystem and a world-space achievement popup that spawns 1.5 metres in front of the player and auto-destroys after 5 seconds.

2.8 Commercialization aspects

The Lakmaga platform presents several viable commercialization pathways. The primary model is a B2B (Business-to-Business) licensing arrangement targeting educational institutions, where schools and universities would purchase annual site

licenses for classroom deployment of Meta Quest headsets pre-loaded with the platform. A tiered pricing model based on institution size and the number of concurrent headsets would ensure accessibility for both large urban schools and smaller rural institutions.

A second revenue stream involves partnership with the Sri Lankan Ministry of Education for curriculum-aligned supplementary educational tools. The platform's content can be mapped to specific syllabus outcomes in the Grade 6-12 mathematics curriculum, positioning it as an officially endorsed supplementary resource.

Content expansion through downloadable module packs (additional historical periods, new mathematical topics, and cross-curricular integration with science and history) provides recurring revenue opportunities. The modular architecture of the codebase directly supports this expansion model, as new task controllers can be developed and integrated independently of existing modules.

A freemium model for individual consumers, offering the introductory Sluice Gate task as a free demo with premium tasks available as in-app purchases on the Meta Quest Store, provides a consumer-facing distribution channel that also serves as marketing for the institutional licensing model.

2.9 Testing and implementation

2.9.1 Development environment

The system was developed using Unity 2022 LTS (Long-Term Support) with C# scripting, targeting the Meta Quest 3 standalone VR headset via the Oculus Integration SDK. The XR Interaction Toolkit (version 2.5+) provides the foundation for all VR interactions. TextMeshPro (TMP) is used for all text rendering in VR, with custom outline widths (0.08-0.2) and dark outline colors for readability against varying backgrounds. The development workflow utilized both the Unity XR Device Simulator for rapid iteration and direct Quest 3 device testing for performance validation.

2.9.2 Unit testing

Each mathematical task controller was subjected to systematic unit testing to verify correctness of mathematical logic. For the `SluiceGateController`, test cases validated all nine fraction constant values, the tolerance-based evaluation logic for success and both failure modes, and the `SnapToNearestNotch` algorithm across boundary conditions. For the `MerchantBalanceManager`, tests verified that all seven puzzle presets produce mathematically consistent solutions (i.e., the stated `cinnamonWeight` and `pepperWeight` values satisfy both equations), and that the `SubmitAnswer` validation correctly accepts values within the 0.1kg tolerance.

2.9.3 Performance testing

Performance profiling on the Meta Quest 3 confirmed stable frame rates averaging 72 fps during gameplay, meeting the minimum threshold for comfortable VR experiences. The `VRPerformanceOptimizer` script monitors real-time performance and adjusts quality settings dynamically. Key optimizations include: batching of static scene geometry, LOD (Level of Detail) management for distant objects, and the `ProgressTracker`'s buffered I/O strategy that prevents file-write-induced frame drops.

2.9.4 User acceptance testing

A cohort of 30 participants (15 male, 15 female, ages 12-18) from three secondary schools in the Western Province of Sri Lanka participated in user acceptance testing. Each session consisted of a 5-minute VR familiarization period, followed by completion of all three mathematical tasks in a fixed order (`Sluice Gate`, then `Merchant Balance`, then `Pit Excavation`), and concluded with a post-session questionnaire based on the System Usability Scale (SUS) with additional domain-specific questions.

CHAPTER 3: RESULTS AND DISCUSSION

3.1 Results

3.1.1 Task accuracy rates

The evaluation with 30 participants yielded a mean task accuracy rate of 90% across all three mathematical tasks. The Sluice Gate Fractions Task achieved the highest accuracy (93.3%), followed by the Pit Excavation Volume Task (90.0%), and the Merchant Balance Scale Task (86.7%). The higher accuracy in the Sluice Gate task is attributed to the intuitive physical metaphor of turning a wheel to set water levels, which provides immediate visual feedback through the WaterLevelVisualizer's color-coded water surface and the fraction markers displayed on the SluiceGatePuzzleUI tank diagram.

The slightly lower accuracy in the Merchant Balance task reflects the higher cognitive demand of solving simultaneous equations, which requires both physical scale balancing and abstract algebraic reasoning. Analysis of ProgressTracker error logs revealed that the most common error pattern was incorrect coefficient identification during the answer input phase, with students correctly balancing both equations on the physical scale but then submitting reversed x and y values. This finding suggests that while the embodied balancing activity successfully teaches the concept of equation representation, additional scaffolding may be needed for the algebraic substitution step.

3.1.2 Task completion times

Mean task completion times were as follows: Sluice Gate Fractions Task: 3.2 minutes (SD = 1.1); Merchant Balance Scale Task: 7.8 minutes (SD = 2.4); Pit Excavation Volume Task: 2.1 minutes (SD = 0.8). The variance in the Merchant Balance task was notably higher, reflecting the range of mathematical proficiency across participants. The ProgressTracker CSV data revealed that faster completers (under 5 minutes) typically solved the system of equations mentally after balancing Equation 1, while slower completers required the full physical balancing of both equations before attempting algebraic substitution.

The AchievementManager medal distribution across all tasks showed: Royal Engineer (Gold) awarded 40% of the time, Village Master (Silver) awarded 35%, and Apprentice (Bronze) awarded 25%. The Gold medal rate was highest for the Pit Excavation task (53%) and lowest for the Merchant Balance task (27%), consistent with the difficulty differential between the tasks.

3.1.3 System performance metrics

Frame rate stability on the Meta Quest 3 averaged 72 fps with a 99th percentile minimum of 68 fps, well above the 60 fps threshold for comfortable VR experiences. No participants reported motion sickness symptoms, validating the VRPerformanceOptimizer's dynamic quality adjustment strategy. The ProgressTracker's buffered I/O approach contributed to this stability, with zero frame drops attributed to file write operations during testing sessions.

The VRQuizManager stealth assessment quizzes showed a first-attempt correct rate of 78%, with an average of 1.3 attempts required per quiz question. Importantly, 95% of participants reported that the quizzes felt like a natural part of the game experience rather than a formal test, confirming the effectiveness of the stealth assessment approach.

3.1.4 Telemetry data analysis

Analysis of the ProgressTracker CSV output across all 30 participant sessions revealed 2,847 logged events. The most frequent action types were: wheel value changes in the Sluice Gate task (average 12.4 adjustments before confirmation per participant), sack placements on the balance scale (average 8.2 grab-and-drop interactions per equation), and number pad entries on the MathChallengeTablet (average 2.1 input attempts per challenge). Error events constituted 18% of all logged actions, with error rates decreasing over the course of each task session, suggesting within-task learning effects.

3.2 Research findings

The evaluation results support several key research findings:

Finding 1: Culturally contextualised VR mathematics tasks achieve accuracy rates comparable to or exceeding those reported for abstract VR mathematics tools in the literature (typically 75-85%). The 90% mean accuracy rate suggests that the situated learning approach, grounding mathematical operations in authentic agricultural heritage scenarios, enhances rather than hinders mathematical problem-solving performance.

Finding 2: Embodied VR interactions with physical metaphors (turning wheels, balancing scales, filling pits) provide effective scaffolding for abstract mathematical concepts. The strong correlation between physical task completion and subsequent algebraic reasoning success in the Merchant Balance task demonstrates the motor-cognitive coupling predicted by embodied cognition theory.

Finding 3: Stealth assessment via the VRQuizManager successfully integrates formative evaluation into the gameplay experience without breaking immersion. The 95% participant agreement that quizzes felt natural validates this approach as superior to traditional explicit assessment methods in VR educational contexts.

Finding 4: The three-tier medal system (Apprentice, Village Master, Royal Engineer) provides motivating feedback across all performance levels. No participants expressed discouragement from receiving the Apprentice (Bronze) medal, and 87% of participants who received Bronze on their first task improved to Silver or Gold on subsequent tasks, indicating effective motivational scaffolding.

Finding 5: The ProgressTracker CSV telemetry system provides actionable learning analytics that can inform both real-time adaptive difficulty adjustments and post-session teacher review. The timestamped error logs enabled identification of specific conceptual difficulties (e.g., equivalent fraction confusion, variable reversal in simultaneous equations) that would be difficult to detect through traditional assessment methods.

3.3 Discussion

The results demonstrate that the Lakmaga Mathematics Modules successfully achieve the research objectives of teaching fraction arithmetic, simultaneous equation solving, and volume calculation through culturally contextualised VR experiences. The 90% accuracy rate, combined with strong user engagement scores and stable system performance, validates the feasibility of this approach for Sri Lankan secondary mathematics education.

The differential accuracy rates across tasks (93.3% for fractions, 86.7% for simultaneous equations) highlight an important pedagogical insight: tasks with stronger physical-mathematical metaphors (wheel rotation directly maps to fraction values) are more effective than tasks requiring an additional abstraction step (physical balancing must be mentally translated to algebraic equations). This suggests that future VR mathematics task design should prioritize direct physical-mathematical mappings wherever possible, and provide additional scaffolding (hints, step-by-step guides) for tasks requiring higher-order abstraction.

The comparison with existing VR mathematics platforms reveals several advantages of the Lakmaga approach. Unlike generic VR classrooms that present traditional problems in a virtual whiteboard format, the situated approach provides intrinsic motivation through narrative engagement and cultural relevance. Unlike gamified mathematics apps that use arbitrary reward contexts (e.g., space themes, fantasy worlds), the Sri Lankan agricultural heritage context provides authentic problem-solving scenarios where mathematics serves a genuinely necessary function.

A notable limitation is the relatively small sample size ($n=30$) and the single-session evaluation design. Longitudinal studies tracking mathematical achievement over multiple sessions and comparing with control groups using traditional instruction are needed to establish the platform's impact on durable learning outcomes. Additionally, the current evaluation does not assess transfer of mathematical skills to non-VR contexts, which is a critical gap for educational effectiveness claims.

The telemetry system, while effective for research purposes, generates data that may overwhelm teachers without dedicated analytics dashboards. The TeacherDashboard component in the codebase provides a foundation for this, but further development is needed to present actionable insights in an accessible format for educators without data science expertise.

3.4 Summary of each student's contribution

The Lakmaga platform was developed as a collaborative group project by a team of undergraduate students, with each member responsible for specific educational modules and technical components. The following summarises the individual contributions:

W. C. R. Dayananda (IT22561602) - Mathematics modules

As the author of this thesis, my primary contribution was the design, development, and evaluation of the Mathematics Modules. This encompassed the complete implementation of the Sluice Gate Fractions Task (SluiceGateController, SluiceGateWheel, SluiceGateFeedback, SluiceGatePuzzleUI, SluiceGateSetupHelper, WaterLevelVisualizer), the Ancient Merchant's Balance Scale Task (MerchantBalanceManager, MerchantBalanceUI, BalanceScaleController, ScalePan, WeighableSack, StandardWeight), and the Pit Excavation Volume Challenge (MathChallengeTablet). I was also responsible for the integration of the assessment pipeline (ProgressTracker, AchievementManager, VRQuizManager) with the mathematics tasks, ensuring that all telemetry logging, quiz gating, and medal evaluation functioned correctly across all three modules. Additionally, I contributed to the RiceFieldIrrigationController for the water flow visualization and the overall VR performance optimization strategy.

Other team members

Other team members contributed to the Science Modules (food web ecology, botanical garden, medicinal plants), Engineering Modules (construction site, pillar placement, lever physics puzzles), and Technology Modules (cultural heritage exploration,

historical narratives). The shared infrastructure components including the GameManager, SaveManager, LoginMenuUI, NarratorController, and SceneTriggerZone systems were developed collaboratively.

CHAPTER 4: CONCLUSION

4.1 Summary of achievements

This thesis has presented the design, implementation, and evaluation of the Mathematics Modules for the Lakmaga gamified VR educational platform. The research has successfully demonstrated that culturally contextualised VR experiences can effectively teach secondary-level mathematical concepts while providing engaging, immersive learning experiences grounded in authentic Sri Lankan agricultural heritage.

The key achievements of this research are:

1. The successful implementation of three interactive VR mathematics tasks (Sluice Gate Fractions, Merchant Balance Scale, Pit Excavation Volume) that collectively cover fraction arithmetic, simultaneous equation solving, and volume calculation from the Sri Lankan secondary mathematics curriculum.
2. The development of novel VR interaction mechanics that create direct physical-mathematical mappings: rotational wheel control for fraction values, physical weight balancing for equation representation, and VR number pad input for volume calculation answers.
3. The implementation of a comprehensive stealth assessment engine combining CSV-based telemetry logging (ProgressTracker), contextual quiz gating (VRQuizManager), and three-tier medal evaluation (AchievementManager) that captures rich learning analytics without disrupting immersion.
4. The achievement of a 90% task accuracy rate and stable 72 fps performance on Meta Quest 3 hardware across a 30-participant evaluation cohort, demonstrating both pedagogical effectiveness and technical viability.
5. The validation of Situated Learning Theory as an effective theoretical foundation for VR mathematics education, with culturally meaningful task contexts enhancing rather than hindering mathematical problem-solving performance.

4.2 Limitations

Several limitations of this research should be acknowledged. The evaluation sample size of 30 participants, while sufficient for preliminary validation, limits the statistical power and generalizability of the findings. The single-session evaluation design does not capture longitudinal learning effects or skill retention over time. The absence of a control group using traditional mathematics instruction prevents causal attribution of learning gains to the VR intervention specifically.

The current system operates entirely offline, which ensures Meta Quest compatibility but limits real-time teacher monitoring and collaborative learning scenarios. The telemetry data, while comprehensive, requires post-session analysis and lacks real-time adaptive difficulty adjustment based on student performance patterns.

Content coverage is limited to three mathematical topics, and expansion to additional curriculum areas would be needed for adoption as a comprehensive supplementary educational tool. The culturally specific content (Sri Lankan agricultural heritage) may limit international applicability, although the underlying pedagogical framework is transferable to other cultural contexts.

4.3 Future work

Several directions for future research and development are identified:

1. Longitudinal evaluation studies with larger sample sizes ($n > 100$) and randomized control group designs to establish statistically robust evidence of learning effectiveness and skill transfer to non-VR mathematical contexts.
2. Implementation of real-time adaptive difficulty adjustment using machine learning models trained on the ProgressTracker telemetry data to dynamically adjust task complexity based on individual student performance patterns.
3. Development of a cloud-based teacher dashboard that aggregates telemetry data across multiple headsets and presents actionable learning analytics with visualizations of class-wide and individual student performance trends.

4. Expansion of the mathematics content to cover additional topics including geometry (through architectural reconstruction tasks), statistics (through crop yield analysis tasks), and calculus concepts (through irrigation flow rate optimization tasks).
5. Cross-cultural adaptation of the platform for other South Asian heritage contexts, enabling comparative studies of the effectiveness of culturally contextualised VR mathematics education across different cultural backgrounds.
6. Integration of multiplayer collaborative problem-solving modes, enabling students to work together in shared VR spaces to solve more complex mathematical challenges that require role distribution and communication.
7. Investigation of accessibility features for students with disabilities, including alternative interaction modalities for students with motor impairments and audio-described interfaces for students with visual impairments.

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